

NARENDRAKUMAR NAGARAJAN

Data Engineering Manager | Enterprise Data Platforms, Fabric & AI

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PROFESSIONAL SUMMARY

Hands on - Data Engineering Manager with **15+ years** of experience in architecting and operating petabyte-scale Lakehouse platforms in Azure. Proven people manager leading cross-functional squads (**10 - 12 engineers**: Data, DevOps, Analytics) to deliver business-critical insights and AI-ready datasets. Deep expertise in **Microsoft Fabric, Databricks, and Azure Data Services**; successfully executed **100+ pipeline migrations** to Fabric with zero data loss, achieving a **40% performance gain**. Delivered **\$390K+/year** in cloud savings through FinOps, drove **100% SFI/GDPR compliance**, and championed **SLO-driven reliability** and "Golden Path" standards to accelerate delivery across high-volume telemetry platforms (**50B+ events/month**).

CORE LEADERSHIP STRENGTHS

- **Data Team Leadership**: Hiring Data Engineers/Analytics Engineers, performance management, and fostering a "DataOps" culture.
- **Data Strategy & Roadmap**: Leading platform modernization (migration to Fabric), capacity planning, and aligning data strategy with business OKRs.
- **Data Governance & Security**: Databricks Unity Catalog, Microsoft Purview, SFI compliance, PII protection, and Zero Trust data access patterns.
- **Operational Excellence**: Managing Data SLAs/SLOs, Data Quality (DQ) frameworks, and incident response for critical pipelines.
- **FinOps & Cloud Efficiency**: Optimizing Spark/Compute clusters, storage lifecycle management, and reducing unit cost of data.

TECHNICAL SKILLS

- **Modern Data Stack**: Microsoft Fabric, Azure Databricks (Spark), Synapse Analytics, ADLS Gen2, Snowflake, Cosmos DB, Logic Apps, Azure Functions, REST APIs, Power BI.
- **Data Engineering**: Python, PySpark, SQL (T-SQL, KQL), C#, Delta Lake, Medallion Architecture (Bronze/Silver/Gold), Event Hubs.
- **GenAI & LLMOps**: Azure AI Foundry, Semantic Kernel, RAG Patterns, Vector Databases, Copilot Studio.
- **DataOps & DevOps**: CI/CD for Data (Azure DevOps), Infrastructure as Code (Terraform), Data Quality Testing.
- **Learning & Experimenting**: Model Context Protocol (MCP), Agent-to-Agent (A2A) Orchestration, Multi-Agent Systems

PROFESSIONAL EXPERIENCE

INFOSYS LIMITED - Technology Lead - CAN | *May 2019 – Present Strategic Partner within Microsoft Engineering, responsible for data engineering leadership, delivery, and platform health.*

Deployed to MICROSOFT - Cyber Defense Engineering | *Sep 2024 – Present Leading the Data Engineering & AI for CDE.*

- **Data Team Leadership**: Manage a cross-functional squad of **10+ engineers** building the security data platform. Improved delivery velocity by implementing **DataOps** best practices (automated testing, schema validation, CI/CD).
- **Strategic Execution**: Own the roadmap for the Security Data Lake, aggregating signals from disparate sources to power threat detection models. Align priorities across Security, Engineering, and Data Science stakeholders.
- **AI-Driven Data Products**: Architected **TICK Agent**, a GenAI-powered data retrieval system on **Azure AI Foundry** that queries massive security logs to automate triage, reducing manual investigation time by **~30%**.
- **Data Governance (SFI)**: Led the **Secure Future Initiative (SFI)** for the data estate; achieved **100% compliance** across 6 critical pillars (Identity, Network, Data Isolation) by hardening storage accounts and access policies.
- **Reliability & Quality**: Established "Golden Path" pipeline standards, resulting in a measurable reduction in data ingestion failures and improved data freshness SLAs.

Deployed to MICROSOFT - Customer Experience & Success | May 2019 – Aug 2024 Led the Modern Support Data Platform, processing enterprise telemetry for Microsoft Knowledge Management Team.

- **Platform Scale:** Owned an enterprise-grade telemetry platform processing **50 Billion+ events/month**. Orchestrated the migration of 100+ pipelines to **Microsoft Fabric** and **OneLake**, achieving a **40% performance gain**.
- **Team Building:** Built and mentored a team of **12 engineers** (Data, Analytics, DevOps). Mentored Senior Data Engineers into Tech Leads and standardized the onboarding process for new hires.
- **FinOps & Cost Optimization:** Reduced Azure data spend by **40% (\$240K/year saved)** by implementing Databricks cluster policies, optimizing Spark shuffle partitions, and utilizing spot instances.
- **DataOps Transformation:** Deprecated manual deployments in favor of fully automated CI/CD pipelines in Azure DevOps, cutting manual release effort by **16 hours/week**.
- **Data Product Delivery:** Partnered with Product to build the data foundation for the **Copilot Recommendation Engine**, delivering curated datasets ("recipes") that improved support agent productivity by **35%**.

INFOSYS LIMITED - Technology Lead | Chennai, India | Aug 2016 – Apr 2019

Deployed to MICROSOFT — Customer Experience & Success

- **Cloud Migration:** Architected the "R3" BI migration from on-prem SQL Server to Azure PaaS (Azure SQL/ADF), reducing annual maintenance costs by **\$150K**.
- **Data Modeling:** Designed dimensional data models (Star Schema) for 150+ entities and delivered 40+ Power BI dashboards used by **500+ Daily Active Users (DAU)**.
- **Data Quality:** Stabilized legacy ETL pipelines, improving Data Freshness SLA adherence from 85% to **99.9%**.

ACCENTURE - Senior Software Engineer | Chennai, India | Feb 2014 – Jul 2016

- **Financial Data Systems:** Designed end-to-end ETL processes using SSIS for royalty calculation and distribution for APRA AMCOS, handling complex business logic for high-volume financial transactions.
- **BI Transformation:** Built a Data Warehouse and BI solution for Stockland Corporation, automating ROI tracking for residential projects and eliminating manual forecasting processes.
- **Automation:** Developed custom automated test data generation tools, reducing QA cycle time by 20%.

CAREVOYANT - Software Engineer Grade 3 | Chennai, India | Dec 2011 – Jan 2014

- **Healthcare Integration:** Developed robust HL7 parsers and SSIS packages to ingest electronic health records (EHR), ensuring accurate data interoperability for US healthcare providers.
- **Backend Development:** Optimized SQL Server stored procedures and WCF services & REST APIs to support high-throughput billing and insurance processing systems

MEDALL HEALTHCARE PVT LTD - Software Engineer | Chennai, India | Sep 2010 - Nov 2011

- **Data Warehousing:** Designed data marts, Data Warehouse, ETL (SSIS) and analytical reports (SSRS) to support revenue forecasting and clinical operations.

EDUCATION & CERTIFICATIONS

Microsoft Certified: Fabric Analytics Engineer Associate (DP-600)

Microsoft Certified: Azure AI Engineer Associate (AI-102)

Microsoft Certified: Azure AI Fundamentals (AI-900)

Bachelor of Engineering (B.E.), Computer Science | Anna University, Chennai, India (2010)